

# Multi-Objective Planning for a Moon Colony: Necessity or Overengineering?

## Summary

Establishing a self-sustaining Moon Colony is a century-scale mega-engineering undertaking with extreme material throughput requirements. Given the project scale and phased nature, we model the colony as a large complex system and separate the problem into two stages: (i) the **construction phase**, which demands massive one-time material delivery, and (ii) the **sustaining phase**, which requires stable, continuous logistics. To ensure consistency and comparability across tasks, we build a unified optimization framework in which each stage is formulated under shared physical, economic, operational, and environmental constraints.

For the construction stage, we propose a hybrid Earth–Moon transportation architecture combining traditional reusable rockets and a **Space Elevator System**. A mixed-integer non-linear programming (MINLP) model allocates annual transport across heterogeneous rocket sites (with latitude-dependent payload correction) and multiple space-elevator hubs, capturing learning effects, fixed operating costs, and throughput ceilings. Under deterministic assumptions, the hybrid solution exhibits a clear baseload–peaking structure: space elevators provide low-marginal-cost baseload throughput, while efficient low-latitude rocket sites supply flexible peaking capacity, enabling project completion in approximately **73–112 years** with a total cost on the order of  $3.4 \times 10^{13}$  **USD**, outperforming both rocket-only (146 years) and elevator-only (186 years) strategies in combined cost–time efficiency.

To address non-ideal operations (Question 2), we incorporate stochastic launch failures, infrastructure disruptions, and macroeconomic budget uncertainty through a penalty-based risk formulation and evaluate system **robustness** using large-scale **Monte Carlo simulation**. Results indicate that single-mode strategies are inherently fragile. Under identical risk assumptions, the hybrid strategy limits expected schedule slippage to approximately **20 years** and incurs a risk premium of about **11%**, achieving an expected completion time of **117.5 years** and an expected total cost of  $3.665 \times 10^{13}$  **USD**, compared with **134.4 years** and  $4.197 \times 10^{13}$  **USD** for the rocket-only alternative.

Beyond large-scale material transport, we extend the framework to the sustaining phase by modeling daily water demand and supply balance within the Moon colony. Population-structured domestic and industrial demands are coupled with recycling, stochastic disturbances, and safety-reserve dispatch, substantially reducing external make-up requirements during peak-demand periods and enhancing the long-term **robustness** of life-support logistics.

Finally, we introduce a **multi-objective decision layer** integrating economic cost, completion time, **robustness**, and environmental impact. Environmental effects are represented through a modular Earth-impact proxy and evaluated within a safety-constrained Pareto framework. A hierarchical optimization perspective coordinates subsystem-level solutions at the system level to produce a practical, resilient, and environmentally responsible strategy, yielding actionable recommendations for the Moon Colony Management (MCM) Agency.

**Keywords:** Space elevator, Monte Carlo simulation, Robustness, Multi-Objective Optimization

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# 1 Introduction



Figure 1: Future Transportation

## 1.1 Problem Background

With the rapid advancement of aerospace technology, human civilization stands at a historic turning point from "space exploration" to "space settlement." According to the strategic planning of the Moon Colony Management (MCM) Agency, humanity will initiate a grand project in 2050: constructing a Moon Colony capable of sustaining a population of 100,000. The core challenge of this vision lies in the immense logistical demand transporting approximately 100 million metric tons of critical materials from Earth to cislunar space.

This scale of transportation places a severe test on traditional logistics paradigms. Traditional chemical propulsion rockets, while technologically mature, are constrained by the Tsiolkovsky rocket equation, resulting in low payload ratios and high marginal costs. Furthermore, high-frequency launches pose potential irreversible threats to Earth's atmospheric environment. In contrast, the Space Elevator System, a revolutionary non-rocket space launch infrastructure, promises low-cost, eco-friendly, and high-throughput access to orbit. However, the construction of a space elevator itself requires a prolonged timeline and massive upfront capital investment. Consequently, determining how to strategically allocate capacity between existing rocket technologies and future space elevators over a century-long horizon balancing total cost, project duration, system robustness, and environmental impact has become a complex systems engineering problem that requires urgent resolution.

## 1.2 Literature Review

Research on Earth-Moon transportation systems has predominantly focused on the physical feasibility and basic economics of individual launch vehicles. regarding traditional transport, early studies focused on orbital mechanics optimization for multi-stage rockets. With the rise of commercial spaceflight, recent literature has shifted attention to the significant impact of reusable launch vehicles (e.g., SpaceX Falcon Heavy) on reducing Low Earth Orbit (LEO) transportation costs [1].

The concept of the space elevator dates back to Konstantin Tsiolkovsky in 1895. The theoretical foundation of the modern space elevator was established by Pearson and others [2]. However, this concept was long constrained by bottlenecks in materials science. In his landmark NIAC study for NASA, Edwards demonstrated that the elevator tether must withstand immense gravitational and centrifugal forces, requiring a material with a specific strength of approximately 30100 GPa [3]. Therefore, the discovery of Carbon Nanotubes (CNTs) marked a pivotal turning point in space elevator research. Studies on the micromechanical properties of CNTs by Iijima and others indicate that their exceptional tensile strength and low density make them the only material theoretically capable of meeting the stringent requirements for

tether construction [4][5].

Despite the extensive literature on the engineering characteristics of rockets and space elevators individually, significant gaps remain. First, most studies treat the two modes as mutually exclusive alternatives rather than complementary components of a unified system. Second, existing economic models often rely on static unit-cost analyses, neglecting the dynamic trade-off between the "time value" of project completion and capacity ramp-up during the construction phase. Finally, few models quantitatively assess system robustness under stochastic risks such as tether severance or launch failures.

To address these gaps, this paper proposes a Mixed-Integer Nonlinear Programming (MINLP) model designed to formulate an optimal long-term logistics strategy for the MCM Agency from a systems engineering perspective.

## 2 Problem Analysis

The transportation system required for the construction and sustained operation of a Moon colony is characterized by tightly coupled constraints across cost, time, risk, and sustainability, which fundamentally limits the applicability of single-metric optimization. Rather than optimizing a single objective, the problem requires balancing multiple competing goals under strict physical and engineering constraints.

A defining difficulty lies in the presence of hard physical limits. Both rocket-based transportation and space elevator systems are constrained by payload capacity, launch or lift frequency, and system throughput, which impose non-negotiable lower bounds on achievable project timelines. Consequently, reductions in completion time cannot be achieved through economic trade-offs alone, and cost and time objectives are only partially substitutable.

When real-world factors such as system failures, maintenance interruptions, and environmental impacts are incorporated, the complexity of the problem increases further. Different transportation modes exhibit distinct risk profiles and asymmetric performance degradation under non-ideal operating conditions, requiring the model to capture not only average efficiency but also robustness under uncertainty.

Our modeling process therefore did not begin with a fully multi-objective formulation. Instead, we initially constructed a cost-minimization model in which the sole objective was to minimize total transportation expenditure subject to demand and capacity constraints. While mathematically well-posed, this formulation produced unrealistically long project completion times, often extending far beyond any plausible strategic or societal planning horizon.

This observation indicated that cost alone is an insufficient criterion for evaluating large-scale interplanetary logistics. We thus extended the model to explicitly incorporate project duration by minimizing a combined cost–time objective, leading naturally to a **mixed-integer nonlinear programming (MINLP)** formulation.

However, further analysis revealed that even a combined cost–time objective remains inadequate to fully represent real-world decision-making. In practice, cost, schedule, system reliability, and long-term sustainability cannot be reduced to a single scalar objective without imposing strong and often unrealistic assumptions about their relative importance.

This insight motivated the adoption of a unified multi-objective decision framework. Rather than prescribing a fixed objective hierarchy, the framework treats different performance dimensions as components of a broader value structure, allowing their relative importance to vary across decision contexts. Within this framework, mixed-integer nonlinear programming serves as the core computational tool for resolving trade-offs in a complex, tightly constrained system.[6]

### 3 Modeling-Mixed-Integer Nonlinear Programming

#### 3.1 Model Overview and System Design

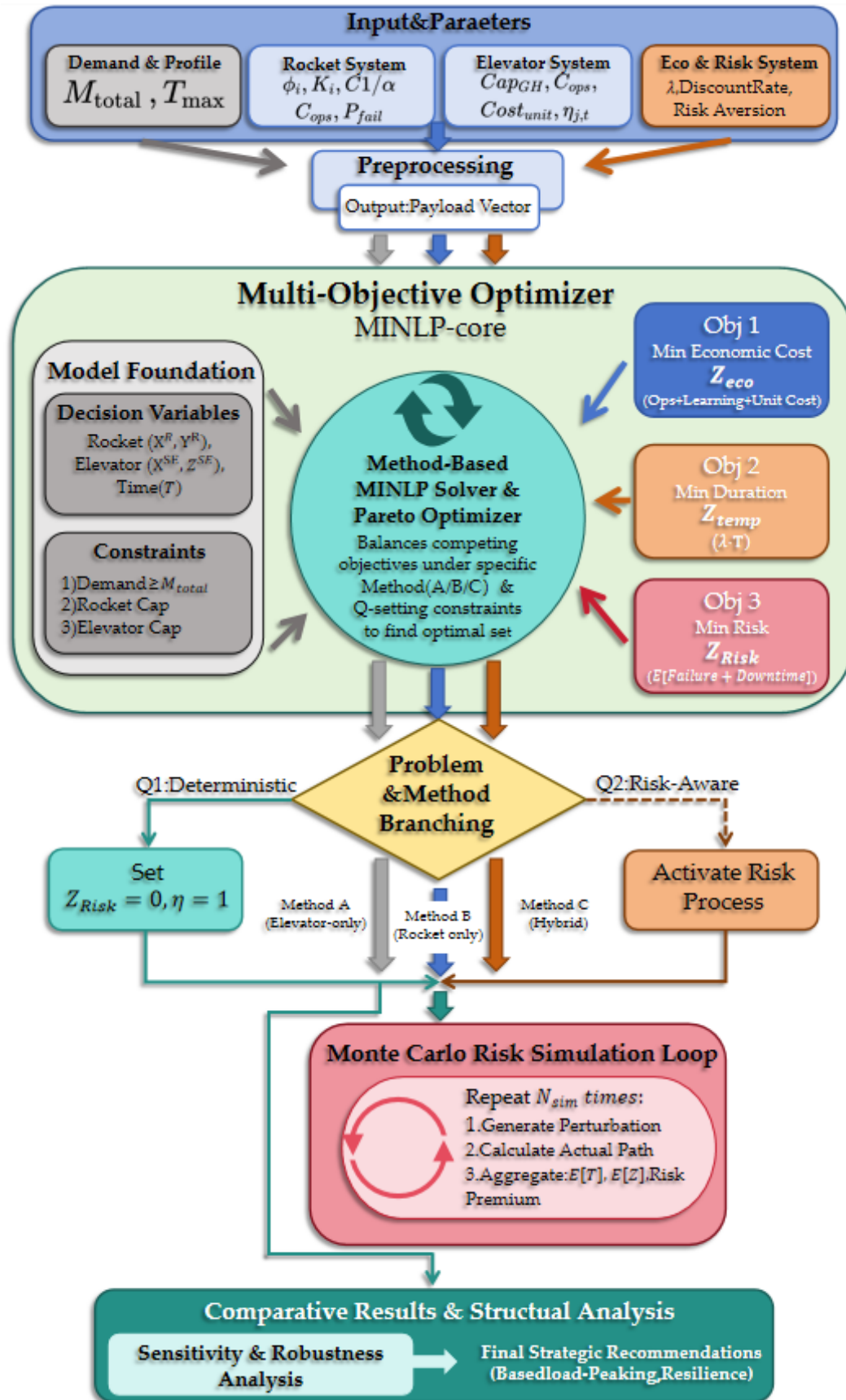


Figure 2: Main Framework

Two fundamentally different transportation modes are considered:

- **Traditional Rocket Transportation (R):** High flexibility and burst capacity, but high marginal cost and failure risk.
- **Space Elevator Transportation (SE):** Extremely low marginal cost but strictly limited throughput.

The transportation system consists of:

- A set of rocket launch sites  $I = \{1, \dots, 10\}$ ;
- A set of equatorial space elevator hubs (Galactic Harbours)  $J = \{1, 2, 3\}$ .

Unlike space elevator hubs, which are assumed to be technologically homogeneous, rocket launch sites exhibit substantial heterogeneity. Even when the same launch vehicle is employed, geographical factors most notably latitude introduce systematic differences in launch efficiency and effective payload capacity.

Consequently, rocket launch sites cannot be treated as interchangeable nodes in the transportation network. Each site must be characterized individually to capture its physical advantages and limitations.



Figure 3: Distribution of Launch Bases

**Latitude-Based Payload Advantage and the Rotation Boost.** Because the initial tangential velocity imparted by Earth's rotation varies with latitude, launch sites at different latitudes exhibit systematic differences in their effective payload capacity for trans-lunar injection. Let  $\phi_i$  denote the geodetic latitude of launch site  $i$ . The local tangential speed contributed by Earth's rotation is approximated by  $v_i = v_{eq} \cos \phi_i$ , where  $v_{eq}$  is the equatorial rotation speed, taken as  $v_{eq} \approx 465$  m/s.

A larger  $v_i$  effectively reduces the required velocity increment ( $\Delta v$ ) provided by the launch vehicle to reach the same target orbit or injection trajectory. Under the differential form of the

Tsiolkovsky rocket equation, small changes in the required  $\Delta v$  translate into approximately linear changes in deliverable payload mass when the vehicle design and staging are fixed. Therefore, for modeling consistency and computational tractability, we approximate the latitude effect on payload using a linear interpolation scheme over the payload range specified in the problem statement for the “Advanced Falcon Heavy,” namely [100, 150] MT.

Specifically, we define the effective payload per launch at site  $i$  as

$$m_i = m_{\min} + (m_{\max} - m_{\min}) \cdot \frac{\cos \phi_i - \cos \phi_{\max}}{\cos \phi_{\min} - \cos \phi_{\max}}, \quad (1)$$

where  $m_{\max} = 150$  MT corresponds to an equatorial launch ( $\phi = 0^\circ$ ), and  $m_{\min} = 100$  MT corresponds to the highest-latitude site among the candidates. This formulation quantitatively captures the low-latitude efficiency advantage and provides a principled basis for differentiating launch sites within the rocket-only and hybrid optimization models.

This formulation quantitatively captures the efficiency advantage of low-latitude launch sites and provides a principled basis for differentiating rocket bases within both the rocket-only and hybrid optimization models.

To ensure an objective comparison across transportation architectures, we quantify two key economic parameters for the space-elevator option: the *lifting cost*  $C_{\text{lift}}$  (Earth surface  $\rightarrow$  apex anchor) and the *apex-to-lunar ferry cost*  $C_{\text{ferry}}$  (apex anchor  $\rightarrow$  Moon). Unlike conventional launch systems constrained by the Tsiolkovsky rocket equation, the space elevator’s operating cost is dominated by electrical energy consumption and routine system maintenance, enabling an order-of-magnitude reduction in marginal cost under mature operations.

**Lifting Cost  $C_{\text{lift}}$ : Parameterization.** Early architectural studies by Bradley Edwards estimate that a first-generation space elevator could achieve an initial operational cost on the order of 220 USD/kg [7]. With technology maturation and networked multi-elevator operations, reports from the International Space Elevator Consortium (ISEC) further anticipate that the long-run transportation cost could fall toward 100 USD/kg or lower [8], approaching the level of electricity cost (around 10 USD/kg) [9] if solar-powered climbers are employed.

**Baseline value.** Considering the assumed 2050-era technology readiness and the scale effects implied by the “Galactic Harbour” system, we set the baseline lifting cost as

$$C_{\text{lift}} = 150 \text{ USD/kg} = 1.5 \times 10^5 \text{ USD/MT}. \quad (2)$$

**Apex-to-Moon Ferry Cost  $C_{\text{ferry}}$ : Parameterization.** The apex anchor is assumed to be located at an altitude of approximately  $1 \times 10^5$  km above Earth’s surface. At this height, the payload inherits a large tangential velocity from Earth’s rotation, which can be sufficient to inject into (or near) a trans-lunar transfer trajectory upon release. Consequently,  $C_{\text{ferry}}$  no longer includes the expensive Earth-escape propellant expenditure typical of ground-launched rockets; it primarily covers (i) terminal braking and soft landing near the lunar surface and (ii) low-intensity trajectory correction and operational support analogous to a space-tug service.

**Baseline value.** Existing research indicates that the energy consumption for deep-space delivery utilizing an apex anchor is extremely low [10]. Because the open literature is comparatively sparse on precise apex-release logistics pricing, we adopt a conservative engineering-economic estimate emphasizing that lunar soft landing still requires chemical propulsion [11]:

$$C_{\text{ferry}} \approx 350 \text{ USD/kg} = 3.5 \times 10^5 \text{ USD/MT}. \quad (3)$$

The total unit transportation cost for the **space elevator** is modeled as

$$Cost_{unit}^{SE} = C_{lift} + C_{ferry}. \quad (4)$$

Under the 2050 baseline values above,

$$Cost_{unit}^{SE} \approx 1.5 \times 10^5 \text{ USD/MT} + 3.5 \times 10^5 \text{ USD/MT} = 5.0 \times 10^5 \text{ USD/MT}. \quad (5)$$

The cost per unit of the **space elevator** has a significant impact on the optimal solution of the final model. We will examine this influence in **Section 4.1**.

The following constraints and objective components are designed to describe the lunar transportation system at a strategic planning level. Rather than modeling individual missions in isolation, the formulation links annual, site-level operational decisions to long-term system feasibility and overall project performance.

Two key principles guide the model construction. First, all transportation modes ultimately serve a single mission objective: satisfying the total material demand required for lunar colonization. Second, different modes exhibit fundamentally different scalability and risk characteristics, which must be reflected explicitly in both constraints and objectives. Together, these principles ensure that short-term operational choices and long-term strategic outcomes are evaluated within a unified decision framework.

### 3.2 Decision Variables and Parameters

The unified model employs the following decision variables:

- $x_{i,t}^R$ : number of rocket launches from site  $i$  in year  $t$ ;
- $y_{i,t}^R \in \{0, 1\}$ : operational status of rocket site  $i$  in year  $t$ ;
- $x_{j,t}^{SE}$ : mass transported by space elevator hub  $j$  in year  $t$  (Unit: MT);
- $z_{j,t}^{SE} \in \{0, 1\}$ : operational status of hub  $j$  in year  $t$ ;
- $T$ : project completion time (Unit: years).

Key system parameters are defined as follows:

- $M_{total}$ : total material demand (Unit: MT);
- $N_{total}$ : Total Number of Launches;
- $m_i$ : effective payload per rocket launch at site  $i$  (latitude-adjusted);
- $K_i$ : annual maximum launch capacity of rocket site  $i$ . To transport **100 million tons** of cargo, the current capacity of **2 or 3 launch pads** per base is far from sufficient. Assuming that by **2050**, as part of a global infrastructure network, each major base is expanded to feature **10 heavy-lift launch bays**, and assuming the turnaround time for a single rocket can be reduced to **3 days**, the calculated value reaches **1,200**;
- $Cap_{GH}$ : annual throughput of one Galactic Harbour (Unit: MT/year);
- $C_{ops}^R, C_{ops}^{SE}$ : annual fixed operating costs, Assuming the values are **\$1 billion/yr** and **\$0.5 billion/yr**, respectively;

- $C_1$ : First-launch baseline cost. Currently, a **Falcon Heavy** launch costs approximately **\$600 million** for a 100-150 ton payload, whereas **Starship** is projected to reduce that to **\$100-\$150 million**. Accounting for technological advancement, we have set the initial launch cost for 2050 at **\$120 million**;<sup>[12]</sup>
- $\alpha$ : rocket learning-curve parameters. According to NASA data, the **cumulative average learning rate** in the aerospace manufacturing industry typically ranges between **85% and 90%**. Given the high repeatability of these missions, a value of **0.85** is applied here<sup>[13]</sup>. To keep the model consistent, we will simultaneously define  $b$  as the **learning rate**.
- $\lambda$ : The **Time Value Coefficient**, represents the opportunity cost or economic loss incurred for every year the project's completion is delayed (Unit: **USD/Year**). For the "2050 Lunar Colony Construction" scenario assumed in this topic, we believe the most appropriate context lies between a **Research-Oriented** approach ( $\lambda = 10$  billion/year) and a **Commercial Strategic** approach ( $\lambda = 50$  billion/year). This range accounts for both the intrinsic value of scientific exploration and the burgeoning potential of the future space economy. For the sake of simplicity, we have set this value at **30 billion/year**.

### 3.3 Constraints and Feasibility

Total transported material from both modes must satisfy mission demand:

$$\sum_{t=1}^T \sum_{i \in I} x_{i,t}^R m_i + \sum_{t=1}^T \sum_{j \in J} x_{j,t}^{SE} \geq M_{\text{total}}. \quad (6)$$

Rocket capacity:

$$x_{i,t}^R \leq K_i y_{i,t}^R, \quad \forall i, t. \quad (7)$$

Space elevator throughput:

$$x_{j,t}^{SE} \leq Cap_{GH} z_{j,t}^{SE}, \quad \forall j, t. \quad (8)$$

Domain constraints:

$$x_{i,t}^R \in \mathbb{Z}_{\geq 0}, \quad x_{j,t}^{SE} \geq 0, \quad y_{i,t}^R, z_{j,t}^{SE} \in \{0, 1\}. \quad (9)$$

### 3.4 Objective Structure and Question-Specific Instantiations

Instead of fixing a single optimization objective, we define a **generalized value function**:

$$\min Z = Z_{\text{economic}} + Z_{\text{temporal}} + Z_{\text{risk}}, \quad (10)$$

where each term represents a different decision dimension.

Economic cost:

$$Z_{\text{economic}} = \sum_{t,i} C_{\text{ops}}^R y_{i,t}^R + \sum_{t,j} C_{\text{ops}}^{SE} z_{j,t}^{SE} + \int_0^{N_{\text{total}}} C_1 n^b dn + \sum_{t,j} x_{j,t}^{SE} \cdot Cost_{\text{unit}}^{SE}. \quad (11)$$

Time-related value:

$$Z_{\text{temporal}} = \lambda \cdot T, \quad (12)$$

In this context, based on **Wright's Cumulative Average Model**, the marginal cost of the  $n$ -th launch,  $MC(n)$ , follows a power-law distribution:

$$MC(n) = C_1 n^b = C_1 n^{\log_2 \alpha} \quad (13)$$

Risk-related penalty:

$$Z_{\text{risk}} = \mathbb{E} [\text{Failure Loss} + \text{Downtime Cost}] . \quad (14)$$

The unified framework is specialized for different questions by selectively activating objective components:

- **Question 1 (Deterministic Planning):**  $Z_{\text{risk}} = 0$ , perfect system reliability assumed.
- **Question 2 (Risk-Aware Planning):**  $Z_{\text{risk}}$  included to account for launch failures and system downtime.

All scenarios (rocket-only, elevator-only, and hybrid systems) are represented as special cases obtained by disabling corresponding decision variables.

This unified formulation reveals that different transportation strategies are not fundamentally different problems, but rather different points within the same decision space. The framework enables consistent comparison across scenarios and provides a systematic foundation for subsequent extensions.

## 4 Results and Comparative Analysis

This section presents the quantitative results and structural interpretation derived from the unified multi-objective decision framework. Results are organized along two dimensions:

- **Operational assumption:** ideal deterministic operation (Question 1) versus non-ideal, risk-aware operation (Question 2);
- **Transportation architecture:** space elevator only, rocket only, and hybrid systems.

Throughout this section, all scenarios are evaluated under a consistent demand requirement of  $M_{\text{total}} = 10^8$  MT.

### 4.1 Deterministic Results (No Failure Assumed)

#### Scenario A: Space Elevator Only

Under deterministic assumptions, the space elevator system is governed by a strict throughput ceiling. With three Galactic Harbours, each providing 179,000 MT/year, the maximum annual capacity is  $R_{\text{SE}} = 537,000$  MT/year

Consequently, the minimum completion time is

$$T_{\text{SE}} = \left\lceil \frac{10^8}{R_{\text{SE}}} \right\rceil \approx 186 \text{ years}. \quad (15)$$

**Analysis.** The space elevator strategy, while providing significant cost efficiency, exhibits strong temporal rigidity, making it suitable only for long-term, low-urgency missions. This rigidity is primarily due to the physical constraints of the system, such as the infrastructure development time, rather than economic factors. Even with ultra-low marginal costs for transporting large quantities of material, the space elevator's throughput is limited by the physical strength of the tether and the operational speed of the climbers, imposing a strict upper bound on its annual capacity. In an ideal scenario, the space elevator could only transport up to 537,000 MT per year, meaning the project would take approximately 186 years to meet the

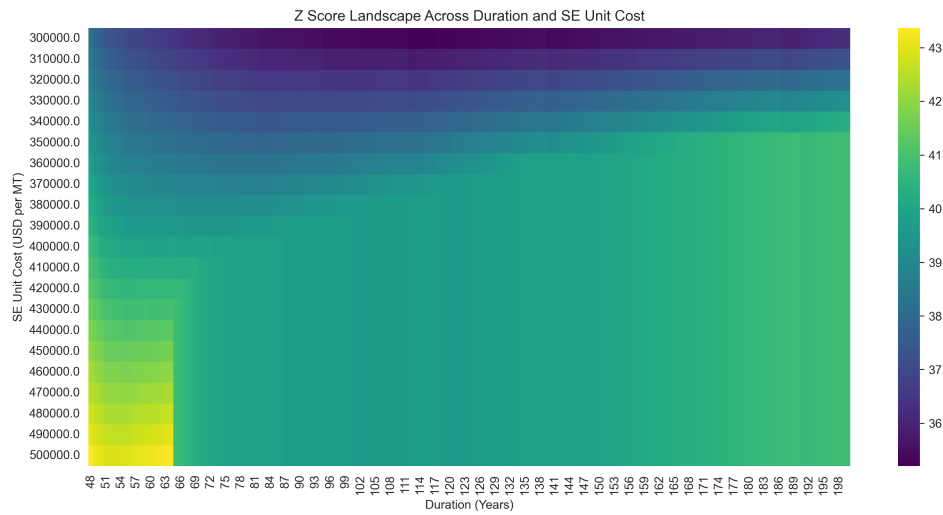


Figure 4: SE cost with Duration

total material demand of  $10^8$  MT. This timeframe, while comparable to the optimal result from the traditional rocket model (146 years), highlights that the space elevator excels in reducing transportation costs rather than speeding up the timeline. Therefore, the space elevator is best suited for missions where cost minimization is prioritized over time urgency, and it becomes a valuable option for long-horizon projects. For scenarios requiring faster completion, the space elevators slow deployment makes it less viable unless combined with faster, higher-throughput systems like traditional rockets.

### Scenario B: Rocket Only

Solving the deterministic MINLP for the rocket-only system yields a clear cost-minimizing solution:

$$T_{\text{opt}} = 146 \text{ years}, \quad Z_{\text{min}} = 3.95 \times 10^{13} \text{ USD}, \quad (16)$$

with a total of  $N_{\text{total}} = 696,849$  launches.

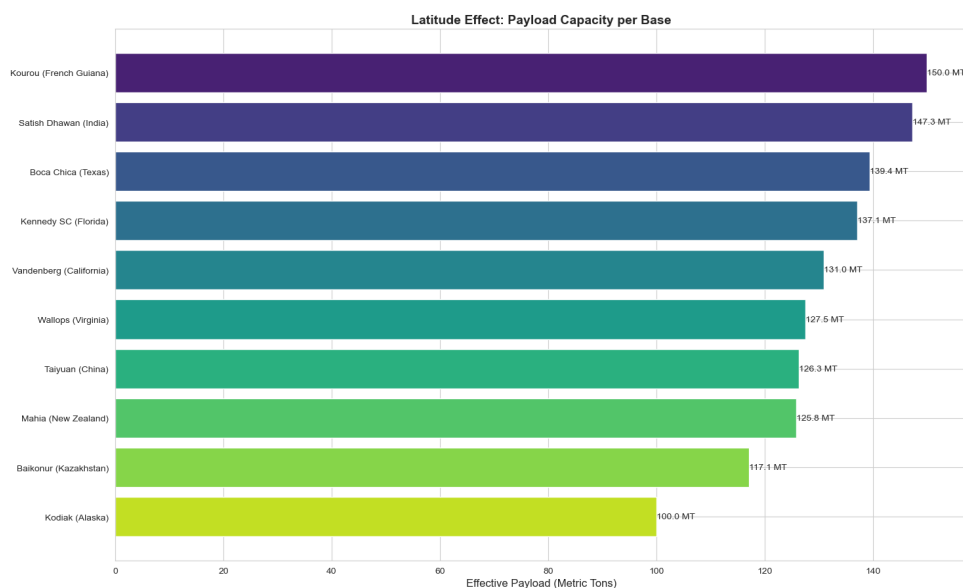


Figure 5: The Maximum Launch Capacity of the Base

Only four low-latitude launch sites are activated, while higher-latitude sites are excluded

due to inferior payload efficiency. Based on the latitude-dependent payload model derived above, the maximum deliverable payload per launch can be evaluated for each candidate launch site. By substituting the geographical latitude of each base into the interpolation formula, we obtain the corresponding upper bound on single-launch transport capacity. The results indicate a clear monotonic relationship between latitude and payload capability: low-latitude sites benefit from stronger Earth-rotation assistance and therefore achieve higher maximum payloads per launch, while high-latitude sites exhibit significantly reduced transport efficiency. These computed site-specific payload limits serve as key input parameters in the subsequent optimization and directly influence launch-site selection and capacity allocation decisions. **Analysis.** The optimization favors an extended construction horizon in order to exploit learning-curve effects and concentrate launches at the most efficient sites. This behavior reveals an implicit assumption of the deterministic model: *time carries no intrinsic penalty*. If society were to care solely about financial expenditure and completely ignore intergenerational or strategic considerations, the optimal solution would indeed converge toward very long timelines.

### Scenario C: Hybrid System (Cost–Time Trade-off)

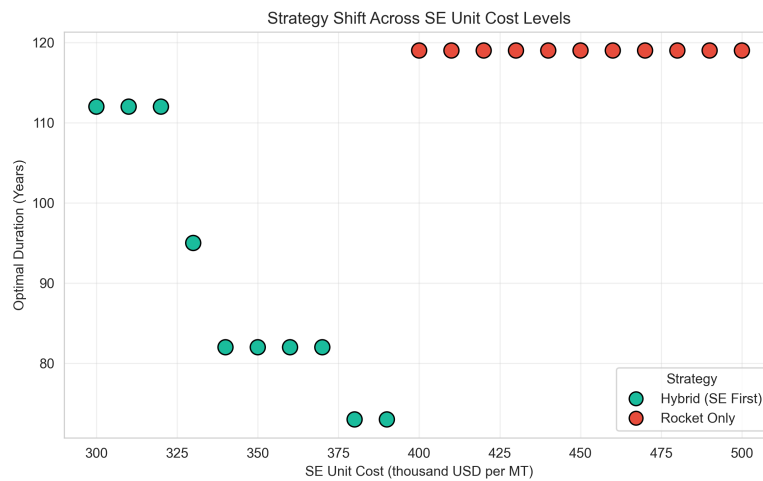


Figure 6: Strategy comparison

When both transportation modes are available, and project duration is explicitly valued, the optimization problem becomes a joint cost–time trade-off. For a given construction horizon  $T$ , the maximum elevator contribution is  $D^{SE}(T) = \min(10^8, 3 \cdot Cap_{GH} \cdot T)$ , with the remaining demand supplied by rockets.

**Analysis.** Analysis of the hybrid model reveals a non-linear evolutionary pattern governed by  $Cost_{unit}^{SE}$ . The optimal strategy shifts through three distinct phases:

- 1. Dominance Phase ( $< 320k$  USD/MT):** With low SE unit costs, the system adopts an "SE-First" strategy (60% volume), stabilizing the optimal duration at **112 years**. The ultra-low marginal costs effectively offset the time penalty of the SEs throughput constraints.
- 2. Acceleration Phase ( $330k \sim 390k$  USD/MT):** Paradoxically, as SE costs rise, the optimal duration **drops sharply to 73 years**. The model shifts toward faster, costlier rockets to minimize the annual  $\lambda = 30$  **billion USD opportunity cost**, demonstrating high strategic flexibility.
- 3. Collapse Point ( $\geq 400k$  USD/MT):** At this "economic lifeline," the SE loses all competitive advantage. The strategy collapses into "**Rocket Only**," with the duration rebounding to 119 years to amortize high launch costs through learning-curve effects.

## 4.2 Risk-Aware Scenario: Non-Ideal Operations and System Uncertainty

In the previous sections, transportation strategies were optimized under idealized deterministic assumptions. However, lunar settlement construction is a mega-engineering project spanning multiple generations and involving global geopolitics as well as frontier technologies. Over such a long construction horizon, the system inevitably faces *technical failures*, *environmental disturbances*, and *socio-political and macroeconomic uncertainty*. Ignoring these non-ideal factors would result in systematically optimistic estimates of both total cost and project duration.

To evaluate the feasibility of transportation strategies under realistic conditions, this section introduces a **stochastic programming** framework and constructs a **generalized robustness model** that quantifies the expected impact of risk on both cost and completion time. Rather than redefining the optimization problem, uncertainty is introduced through a **penalty-based formulation**, ensuring consistency with the deterministic framework. Stochastic events affect system performance by reducing delivered payload, degrading available capacity, and indirectly extending the project duration.

System performance under uncertainty is evaluated using a **Discrete Event Monte Carlo (DEMC)** simulation framework, which captures the cumulative effects of failures, environmental perturbations, and budget constraints over the full project horizon.

### 4.2.1 Risk and Uncertainty Modeling

**Rocket transportation system.** Despite significant improvements in reliability by 2050, conventional rocket transportation cannot fully eliminate catastrophic failures due to the inherent nature of high-energy chemical propulsion. We assume that each rocket launch fails independently with probability

$$P_{\text{fail}}^R = 0.01.$$

When a failure occurs, the system suffers not only the loss of launch cost, but also the destruction of valuable payload. Let  $V_{\text{payload}}$  denote the average economic value of transported material. The risk-adjusted expected cost of a single launch is expressed as

$$\mathbb{E}[C_{\text{launch}}] = (1 - P_{\text{fail}}^R) C_{\text{marginal}}^R + P_{\text{fail}}^R \left( C_{\text{marginal}}^R + m_i V_{\text{payload}} + C_{\text{investigation}} \right), \quad (17)$$

where  $C_{\text{investigation}}$  represents hidden costs associated with accident investigation and safety review.

More importantly, a launch failure triggers a *shutdown fuse* mechanism at the affected site. During the subsequent repair period, the site is forced into a non-operational state, reducing its effective annual capacity. This stochastic capacity loss may extend the overall project duration, thereby amplifying total cost through accumulated fixed operating expenses and time opportunity cost.

### Space elevator system.

The space elevator exhibits a mixed risk profile characterized by the coexistence of frequent low-impact disturbances and rare high-impact catastrophic events.

Environmental perturbations such as solar wind, gravitational disturbances, and Coriolis-induced effects cause nonlinear lateral tether sway, reducing effective throughput. This effect is captured by a tether stability factor  $\eta_{\text{sway}} \in (0, 1]$ , which represents the fraction of nominal throughput that can be reliably achieved. As a result, the effective annual capacity of a Galactic Harbour is given by

$$Cap_{\text{eff}}^{SE} = Cap_{GH} \cdot \eta_{\text{sway}}.$$

In addition, catastrophic tether severance events may occur with annual probability

$$P_{\text{sever}} = 0.001.$$

Once severance occurs, the affected harbour is completely shut down for a reconstruction period of

$$T_{\text{rebuild}} = 5 \text{ years},$$

during which its transport capacity is zero. In the absence of backup capacity, the system must rely on additional rocket launches to fill the resulting throughput gap, leading to sharp cost fluctuations.

### Socio-political and budget uncertainty.

In addition to physical failures, long-term space infrastructure projects are highly sensitive to political and economic conditions. We therefore introduce stochastic macro-level constraints to represent geopolitical disruptions and uncertainty in budget approval.

Regional conflicts or diplomatic disruptions may render individual rocket sites temporarily unavailable, forcing transportation demand to be reallocated to other sites and potentially increasing marginal cost. At the same time, project funding is subject to political approval cycles. We model this effect using a stochastic annual budget cap

$$B_{\text{max}} = 100 \text{ billion USD.}$$

When planned expenditure exceeds the approved budget, transportation activities must be curtailed, reducing annual throughput and indirectly extending the project duration. Such budget-induced delays accumulate fixed operating costs over time and constitute a major source of systemic risk in long-horizon projects.

### 4.2.2 Robustness Results and Strategy Comparison

To quantitatively evaluate the cumulative impact of non-ideal operating conditions over the entire construction horizon, we adopt a **Discrete Event Monte Carlo (DEMC)** simulation framework. Rather than relying on closed-form analytical solutions, the transportation system is modeled as a stochastic dynamic process in which technical failures, environmental disturbances, and macro-level constraints jointly drive the time evolution of cost and progress.

The continuous project timeline is discretized into uniform time steps with interval  $\Delta t$  (set to one month). At time step  $t$ , the system state is characterized by the availability of rocket launch sites, the effective throughput of space elevator harbours, and the approved annual budget. Stochastic disturbance realizations are generated according to the prescribed risk models, and the effective system transport capacity  $Cap_{\text{sys}}(t)$  is determined accordingly.

Let  $M_{\text{rem}}(t)$  denote the remaining material demand at time  $t$ . When  $M_{\text{rem}}(t) > 0$ , a local constrained allocation problem is solved to determine the transport decision for the current period:

$$\mathbf{x}_t^* = \arg \min_{\mathbf{x}_t} \left( \sum_{i \in I} x_{i,t}^R C_{\text{marginal}}^R(t) + \sum_{j \in J} x_{j,t}^{SE} C_{\text{unit}}^{SE} \right), \quad (18)$$

subject to the physical capacity constraint

$$\sum_{i \in I} x_{i,t}^R + \sum_{j \in J} x_{j,t}^{SE} \leq Cap_{\text{sys}}(t), \quad (19)$$

and the stochastic budget constraint

$$\sum_{i \in I} x_{i,t}^R C_{\text{marginal}}^R(t) + \sum_{j \in J} x_{j,t}^{SE} C_{\text{unit}}^{SE} \leq B_{\text{max}}(t). \quad (20)$$

If the realized capacity or budget is insufficient, the untransported material is carried forward, resulting in endogenous schedule slippage. The remaining demand is updated as

$$M_{\text{rem}}(t + \Delta t) = M_{\text{rem}}(t) - \left( \sum_{i \in I} x_{i,t}^R + \sum_{j \in J} x_{j,t}^{SE} \right). \quad (21)$$

Transportation cost and delivered mass are accumulated over successive time steps until  $M_{\text{rem}}(t) \leq 0$ . The completion time  $T_k$  and total cost  $Z_k$  of the  $k$ -th realization are then recorded. Repeating this process for  $N = 10,000$  independent Monte Carlo runs yields empirical distributions of project duration  $T$  and total cost  $Z$ , which serve as the basis for robustness evaluation under non-ideal operating conditions.

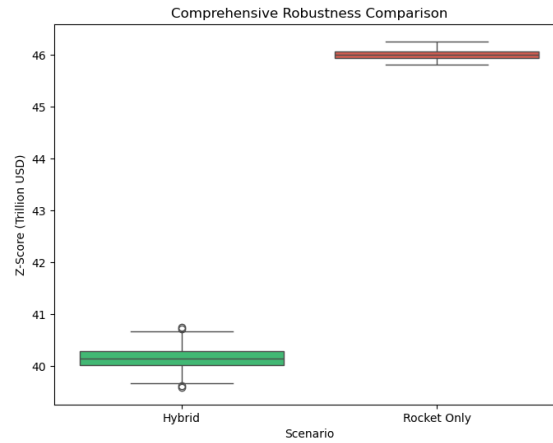
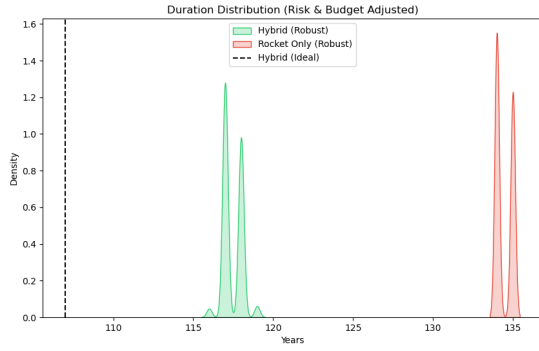


Figure 7: Robustness comparison of single-mode transportation systems under non-ideal operating conditions.

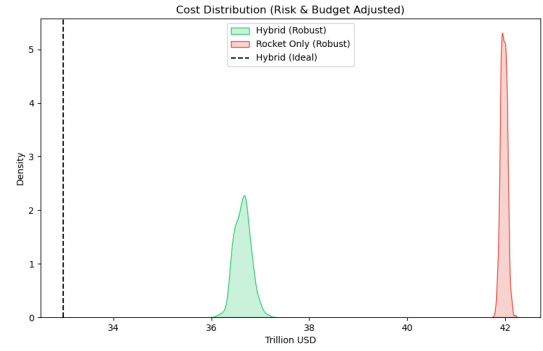
Introducing risk fundamentally changes the evaluation criterion of transportation strategies. Solutions that are optimal in deterministic settings may become fragile once exposed to repeated disruptions.

- **Rocket-only systems.** Although flexible under ideal conditions, rocket-only transportation requires nearly  $7 \times 10^5$  launches. Even a small per-launch failure probability accumulates into a large expected number of failures. Each failure induces payload loss and site downtime, reducing effective capacity and forcing additional launches later, creating a self-reinforcing cycle of delay and cost escalation.
- **Space-elevator-only systems.** Space elevators are not dominated by frequent discrete events, but suffer from strong single-point dependency. While throughput degradation due to tether sway is manageable, rare severance events impose multi-year shutdowns with no substitute capacity, resulting in pronounced tail risk.

By integrating rockets and space elevators, the hybrid system introduces structural redundancy that allows one mode to compensate for disruptions in the other. Under the same risk parameterization, Monte Carlo simulation yields



(a) Expected completion time under risk



(b) Expected total cost under risk

Figure 8: Comparison between rocket-only and hybrid transportation systems under non-ideal operating conditions.

$$\mathbb{E}[T_{\text{hybrid}}] = 117.5 \text{ years}, \quad \mathbb{E}[Z_{\text{hybrid}}] = 3.665 \times 10^{13} \text{ USD}. \quad (22)$$

Relative to the idealized deterministic baseline, this corresponds to an average delay of approximately 20 years and a risk premium of about 11%. The additional cost is primarily driven by high-value payload loss from rocket failures and substitution costs when elevator throughput is temporarily degraded.

Despite these penalties, the hybrid system remains significantly more robust than the rocket-only alternative, which exhibits an expected completion time of 134.4 years and an expected total cost of  $4.197 \times 10^{13}$  USD under the same risk assumptions.

## 5 Water-Demand Modeling and Daily Supply Balance

### 5.1 Model Formulation and Pipeline

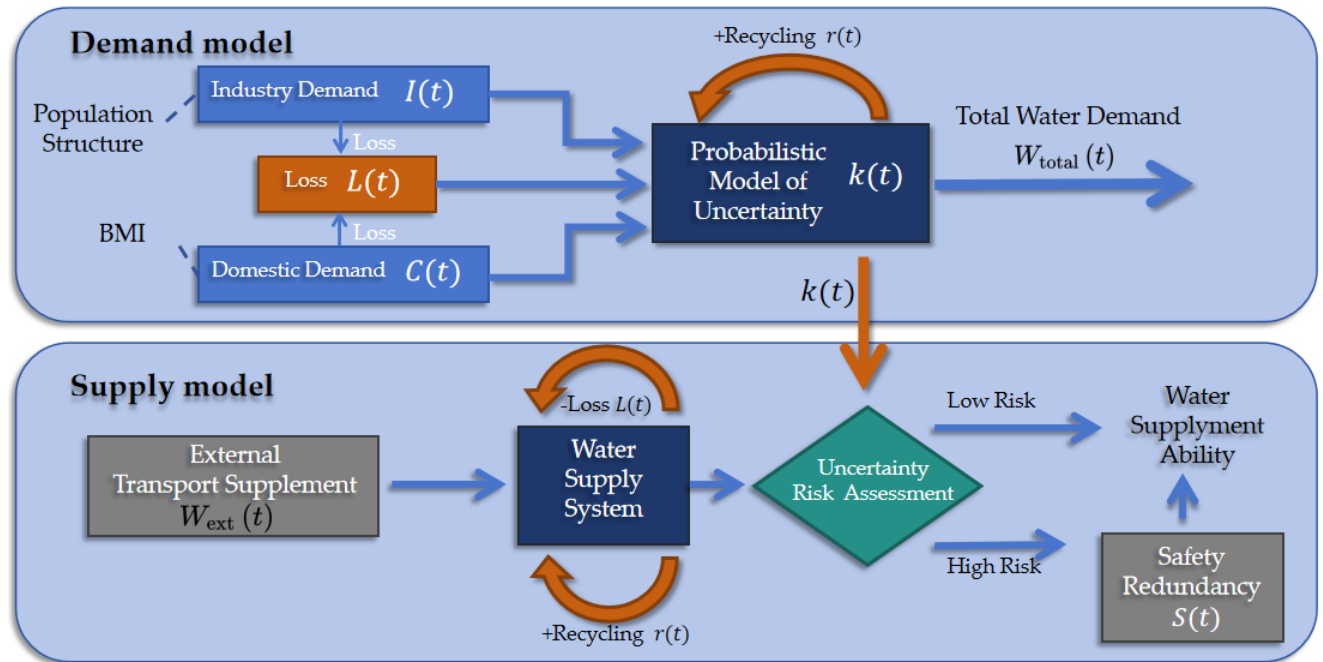


Figure 9: Pipeline of Problem 3

Figure 9 summarizes our modeling pipeline: we first compute population-structured per-capita domestic and industrial demands, then generate a daily disturbance multiplier, and finally close the daily mass balance through recycling, losses, external make-up, and safety-reserve dispatch.

We model the system on a daily time scale. Let  $t \in \{1, \dots, T\}$  denote the day index within one operational year ( $T = 365$ ). The model outputs daily gross demands  $w_L(t)$  and  $w_I(t)$  (ton/day), their total  $w_{\text{total}}(t)$ , and the net external make-up requirement  $w_{\text{ext}}(t)$  (ton/day). Annual totals are obtained by aggregation:

$$W_{\text{total}} = \sum_{t=1}^T w_{\text{total}}(t), \quad W_{\text{ext}} = \sum_{t=1}^T w_{\text{ext}}(t). \quad (23)$$

We use  $1000 \text{ L} \approx 1 \text{ m}^3 \approx 1 \text{ ton}$  for unit conversion.

Let  $N(t)$  be the population on day  $t$ . We partition the population by the same age groups  $g \in \mathcal{G}$  and sex  $s \in \{m, f\}$  for both domestic and industrial submodules. Denote group counts by  $N_{g,s}(t)$  and weights by

$$w_{g,s}(t) = \frac{N_{g,s}(t)}{N(t)}. \quad (24)$$

Domestic demand  $C(t)$  (age–sex–BMI structured). The domestic submodule computes group-level BMI summaries  $\text{BMI}_{g,s}(t)$  and maps them to per-capita daily domestic demand via an interpretable saturating response. The population-structured per-capita domestic demand is

$$C(t) = \sum_{g \in \mathcal{G}} \sum_{s \in \{m, f\}} w_{g,s}(t) c_{g,s}(\text{BMI}_{g,s}(t)), \quad (25)$$

where  $c_{g,s}(\cdot)$  is defined in the domestic submodule.

Industrial demand  $I(t)$  (age-structure nonlinearity). The industrial submodule uses the same weights  $w_{g,s}(t)$  to compute age-structure statistics (e.g., moments/dispersion and sex fraction) and returns the per-capita industrial demand  $I(t)$  in L/(person · day).

To capture daily surges and persistence, we introduce a disturbance multiplier  $k_t \geq 1$  generated by an interpretable regime-switching stochastic process (defined in the disturbance submodel). If different sensitivities are desired, we use

$$k_{L,t} = 1 + \eta_L(k_t - 1), \quad k_{I,t} = 1 + \eta_I(k_t - 1). \quad (26)$$

Daily gross demands (ton/day) are then

$$w_L(t) = k_{L,t} \cdot \frac{N(t) C(t)}{1000}, \quad w_I(t) = k_{I,t} \cdot \frac{N(t) I(t)}{1000}, \quad w_{\text{total}}(t) = w_L(t) + w_I(t). \quad (27)$$

The supply side consists of recycling, unavoidable losses, and external make-up delivered by transportation. Let  $r_L$  and  $r_I$  be effective recycling ratios. The recycled amount is

$$r(t) = r_L w_L(t) + r_I w_I(t), \quad (28)$$

and the net consumption after recycling is

$$w_{\text{net}}(t) = (1 - r_L)w_L(t) + (1 - r_I)w_I(t). \quad (29)$$

Losses are modeled by an affine form

$$\ell(t) = \lambda_d w_{\text{total}}(t) + \ell_0. \quad (30)$$

Finally, the net external make-up requirement is

$$w_{\text{ext}}(t) = w_{\text{net}}(t) + \ell(t) + \Delta s(t), \quad (31)$$

where  $\Delta s(t)$  represents planned inventory build-up (if any).

We introduce a dedicated safety reserve that is dispatched only under elevated operational risk. The risk trigger is extracted directly from the disturbance process. Specifically, we set a surge threshold as an upper quantile of  $\{k_t\}$ :

$$k^{\text{thr}} = \text{Quantile}_{\alpha}(\{k_t\}_{t=1}^T), \quad \alpha \in (0.9, 0.99), \quad (32)$$

and define the risk indicator  $\mathcal{I}_t = \mathbf{1}[k_t \geq k^{\text{thr}}]$  (or conservatively  $\mathbf{1}[S_t = E]$  when the regime state is used). The reserve is dispatched only when  $\mathcal{I}_t = 1$  to cover the shortfall between required make-up and planned transport supply:

$$v_{\text{safe}}(t) = \mathcal{I}_t \cdot \min\{S_{\text{safe}}(t-1), [w_{\text{ext}}(t) - v_{\text{trans}}(t)]^+\}, \quad [x]^+ = \max\{x, 0\}. \quad (33)$$

## 5.2 Result Analysis for Problem 3

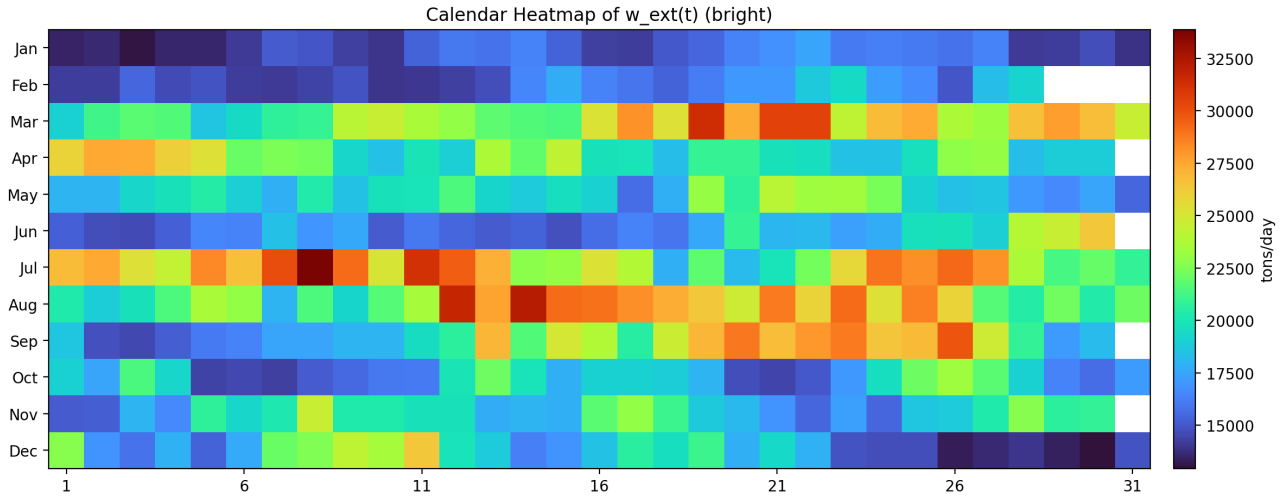


Figure 10: Calendar heatmap of  $w_{\text{ext}}(t)$  (month  $\times$  day).

The calendar heatmap reveals a clear seasonal pattern in the external make-up requirement  $w_{\text{ext}}(t)$ : extended high-demand clusters concentrate in mid-year (notably around March and July–September), whereas early-year and late-year days remain comparatively low. More importantly, peak periods appear as *multi-day clusters* rather than isolated spikes, implying persistence in the disturbance process. This persistence motivates (i) sizing transport make-up capacity using upper-quantile days rather than daily averages, and (ii) introducing a safety reserve that can cover several consecutive high-demand days.

Figure 11 shows that the gross demand  $w_{\text{total}}(t)$  varies strongly over the year, while the required external make-up  $w_{\text{ext}}(t)$  is substantially lower. In our baseline results, the **average daily external make-up demand is about 50,000 MT/day**, whereas the planned external supply capability is **about 100,000 MT/day**, leaving a margin to absorb routine fluctuations. Figure 12 explains this gap: recycling  $r(t)$  scales with demand and smooths peaks, while the loss term  $\ell(t)$  behaves as a relatively stable overhead. Overall, recycling acts as an endogenous peak-shaving mechanism, reducing both the mean and variability of  $w_{\text{ext}}(t)$ .

The distribution of the disturbance multiplier  $k_t$  exhibits a pronounced right tail, implying that rare but persistent surges drive heavy-tailed  $w_{\text{ext}}(t)$ . Figure 14 shows that reserve

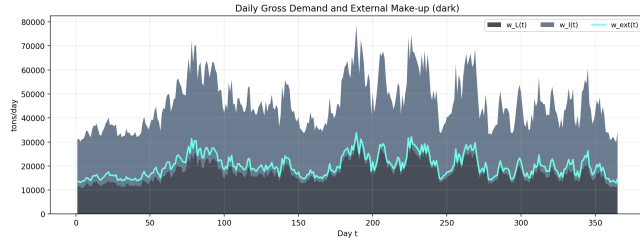


Figure 11: Daily gross demand components  $w_L(t)$ ,  $w_I(t)$  and external make-up  $w_{\text{ext}}(t)$ .

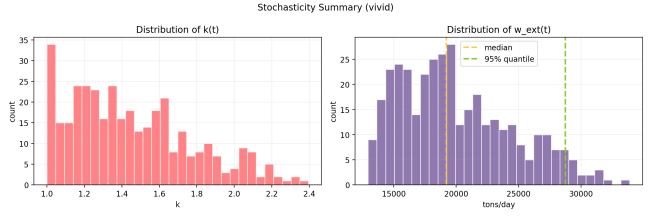


Figure 12: Supply-side decomposition: recycled  $r(t)$ , loss  $\ell(t)$ , net-after-recycle  $w_{\text{net}}(t)$ , and  $w_{\text{ext}}(t)$ .

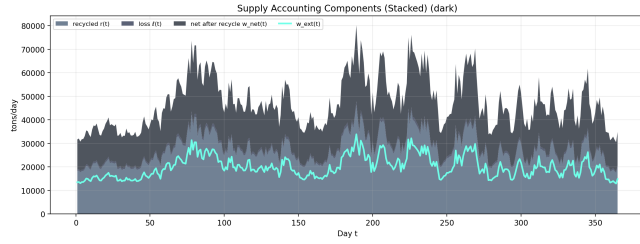


Figure 13: Distributions of  $k_t$  and  $w_{\text{ext}}(t)$  (with median and 95% quantile markers).

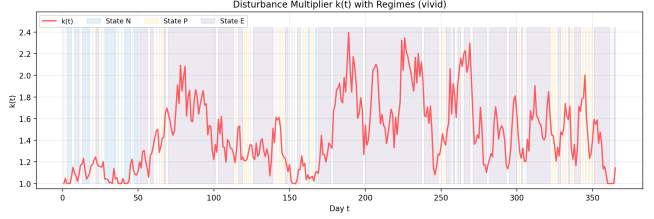


Figure 14: Safety reserve dynamics: inventory  $S_{\text{safe}}(t)$  and dispatch  $v_{\text{safe}}(t)$ .

dispatch  $v_{\text{safe}}(t)$  is activated intermittently and concentrates in short bursts, consistent with tail-risk days. To ensure continuity under clustered surges, we size the safety reserve at **500,000 MT**, which is sufficient to cover roughly **10 days** of average external make-up demand ( $\approx 50,000$  MT/day), or several consecutive peak days when transport is temporarily constrained. The inventory  $S_{\text{safe}}(t)$  decreases during shock periods and replenishes afterward, indicating that the reserve functions as a targeted buffer for extreme events rather than a continuously consumed source.

## 6 Transportation Planning and Earth-Impact Proxy Model

### 6.1 Model Setup and Earth-Impact Proxy

Figure 15 summarizes our pipeline: annual payload demand  $M(t)$  is formed by combining water demand  $W_{\text{ext}}(t)$  with other payloads, then allocated between rockets and the space elevator under capacity constraints, and finally evaluated by an economic objective and a modular impact proxy.

Let  $t = 1, 2, \dots, T$  denote the planning period (interpreted as years when coupled with the daily water model). The annual payload demand is

$$M(t) = M_{\text{water}}(t) + M_{\text{other}}(t), \quad M_{\text{water}}(t) := W_{\text{ext}}(t).$$

Decision variables are the number of rocket launches  $N_R(t) \in \mathbb{Z}_{\geq 0}$  and elevator-delivered mass  $D_{SE}(t) \geq 0$ . The mass balance and elevator capacity constraints are

$$m_R N_R(t) + D_{SE}(t) \geq M(t), \quad 0 \leq D_{SE}(t) \leq \overline{D}_{SE}(t). \quad (34)$$

Total economic cost over the horizon is

$$Z_{\text{econ}} = C_{SE}^{\text{build}} + \sum_{t=1}^T \left( C_R(N_R(t)) + C_{SE}^{\text{op}}(D_{SE}(t)) \right). \quad (35)$$

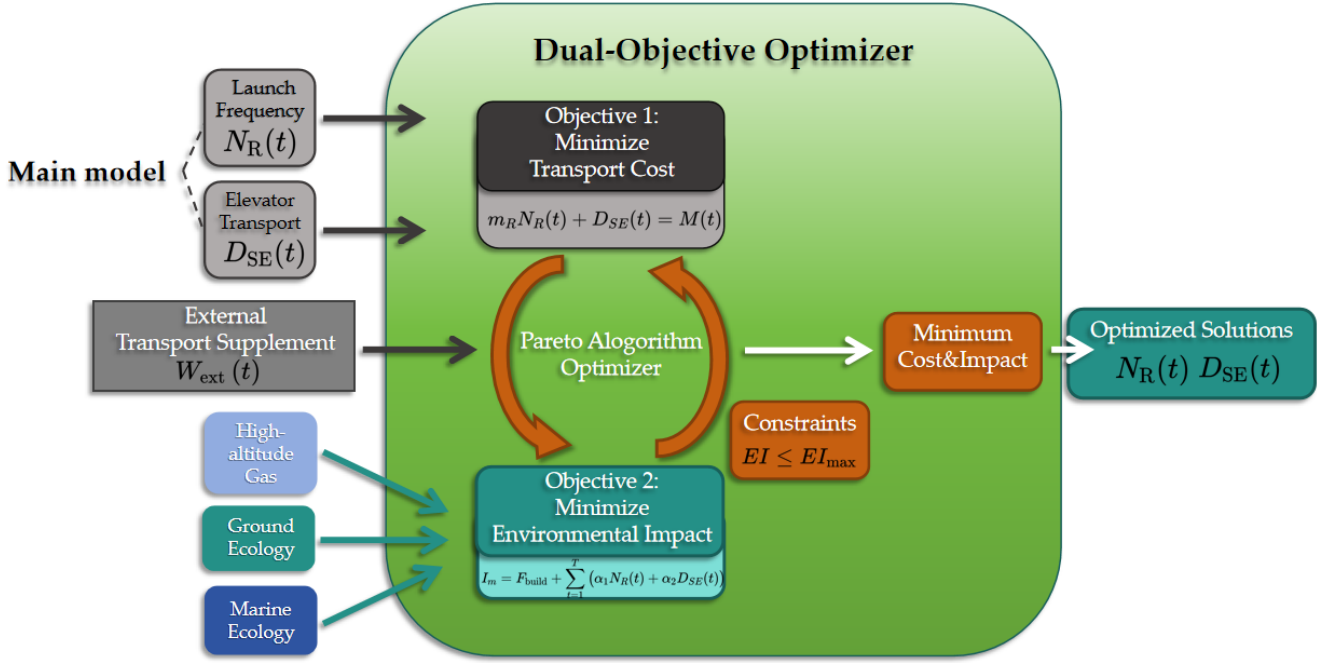


Figure 15: Pipeline of Problem 4

To keep the model interpretable and data-light, we represent Earth impacts through a small set of modules  $\mathcal{M}$  (e.g., atmospheric, ecological, life-cycle, and risk). For each module  $m \in \mathcal{M}$ , we use a fixed-plus-marginal structure:

$$I_m = F_{m,SE}^{\text{build}} + \sum_{t=1}^T (\alpha_{m,R} N_R(t) + \alpha_{m,SE} D_{SE}(t)). \quad (36)$$

A normalized composite index is then computed as

$$EI = \left( \sum_{m \in \mathcal{M}} \left( \frac{I_m}{\bar{I}_m} \right)^2 \right)^{1/2}, \quad EI \leq EI_{\max}. \quad (37)$$

We treat  $(Z_{\text{econ}}, EI)$  as a bi-objective system under the hard safety cap  $EI \leq EI_{\max}$ . The safe Pareto set can be generated by an  $\varepsilon$ -constraint sweep, and a unique compromise can be selected using a bargaining criterion (details omitted here for brevity).

**Decision:**  $\{N_R(t) \in \mathbb{Z}_{\geq 0}, D_{SE}(t) \in \mathbb{R}_{\geq 0}\}_{t=1}^T$ .

**Constraints:**  $m_R N_R(t) + D_{SE}(t) \geq M(t), \quad 0 \leq D_{SE}(t) \leq \bar{D}_{SE}(t), \quad \forall t,$

$$I_m = F_{m,SE}^{\text{build}} + \sum_{t=1}^T (\alpha_{m,R} N_R(t) + \alpha_{m,SE} D_{SE}(t)), \quad \forall m \in \mathcal{M},$$

$$EI = \left( \sum_{m \in \mathcal{M}} (I_m / \bar{I}_m)^2 \right)^{1/2} \leq EI_{\max}.$$

**Economy:**  $Z_{\text{econ}} = C_{SE}^{\text{build}} + \sum_{t=1}^T (C_R(N_R(t)) + C_{SE}^{\text{op}}(D_{SE}(t))).$

## 6.2 Result Analysis for Problem 4

**Analysis.** Figure 16 exhibits a clear economy–environment tradeoff: as  $Z_{\text{econ}}$  decreases,  $EI$  tends to increase, forming a well-structured safe Pareto frontier under the constraint  $EI \leq$

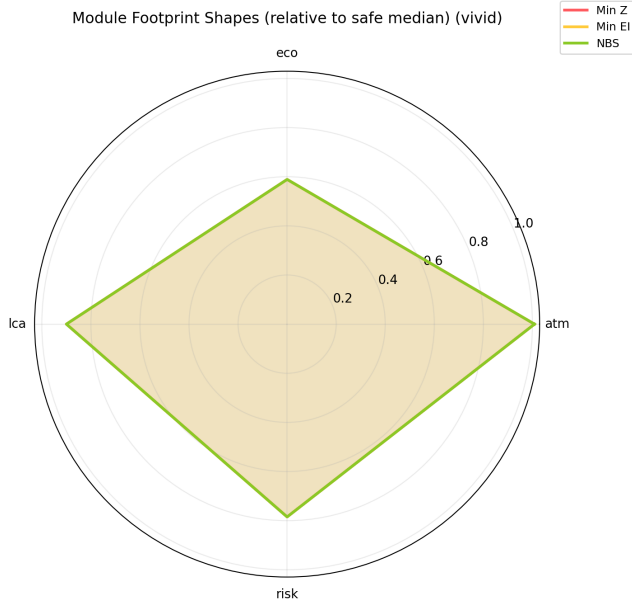


Figure 16: Objective space ( $Z_{econ}, EI$ ) with safe set ( $EI \leq EI_{max}$ ), safe Pareto frontier, and Nash bargaining solution (NBS).

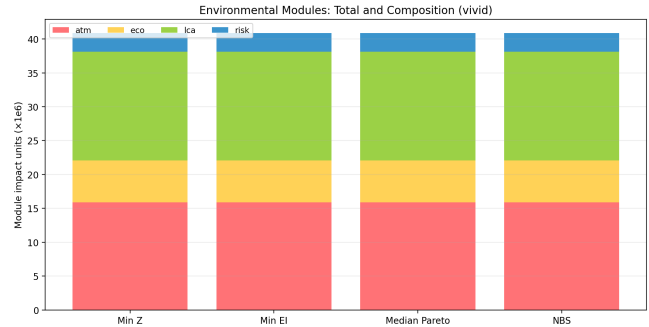


Figure 17: NBS transport mix over time: rocket launches  $N_R(t)$  and elevator throughput  $D_{SE}(t)$ .

$EI_{max}$ . The horizontal bound  $EI_{max}$  truncates high-impact candidates, leaving a feasible “safe” region in which the frontier represents the best attainable compromises. The NBS point lies on (or extremely close to) the safe Pareto frontier and remains strictly below  $EI_{max}$ , indicating that the bargaining-based selection avoids dominated compromises while maintaining a safety margin in environmental impact. Figure 17 shows the resulting time-phased allocation:  $N_R(t)$  stays relatively stable across years, whereas  $D_{SE}(t)$  varies substantially, implying that the elevator channel is used opportunistically when its effective capacity is available, while rockets provide a steady baseline to guarantee feasibility under capacity and mass-balance constraints.

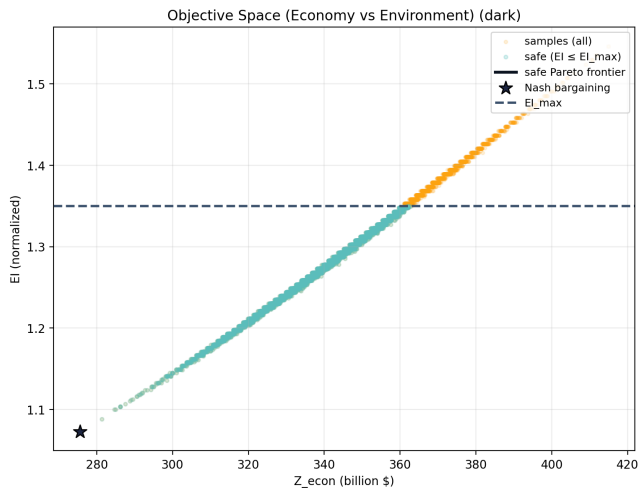


Figure 18: Environmental modules: total impact and composition for representative solutions (Min  $Z$ , Min  $EI$ , Median Pareto, NBS).

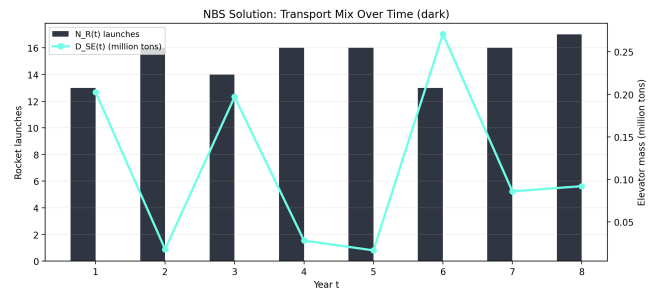


Figure 19: Module footprint shapes (normalized): comparison among Min  $Z$ , Min  $EI$ , and NBS.

**Analysis.** Figure 18 decomposes the overall environmental metric into module impacts  $I_m$  and shows that the total impact is dominated by a subset of modules (notably the larger stacked components), while the remaining module(s) contribute comparatively less. Across

representative solutions (including NBS), the *composition* is broadly consistent, suggesting that optimization within the safe region primarily adjusts the overall scale of impact rather than completely shifting the burden among modules. Figure 19 reinforces this observation in a shape-oriented view: the normalized footprint of NBS is comparatively smooth and close to the other representative solutions, indicating that the bargaining compromise avoids extreme “single-module spikes” and yields a balanced environmental profile. Consequently, further reduction in *EI* is more likely to require structural improvements in the dominant module drivers (e.g., lowering the high-contributing module coefficients or improving their corresponding technologies), rather than relying solely on reallocation among existing transport decisions.

## 7 Letter of Recommendation to the Moon Colony Management (MCM) Agency

Dear Members of the Moon Colony Management Agency,

The construction and long-term operation of a Moon colony capable of supporting a population of 100,000 people represents a project of unprecedented scale and strategic significance. Such an undertaking must be viewed as a century-scale systems engineering challenge characterized by long operational horizons, cumulative uncertainty, and multiple competing objectives. Transportation decisions made in the early stages will have persistent impacts on cost, schedule, system resilience, resource security, and environmental sustainability. Based on our integrated modeling and analysis, we respectfully submit the following recommendation for your consideration.

Our study indicates that a hybrid transportation architecture combining a Space Elevator System with traditional rocket launches provides the most balanced and robust solution for both the construction phase and long-term operation of the Moon colony. In this framework, the space elevator serves as the backbone of stable, low-cost, and environmentally sustainable baseline transport capacity, while rockets provide flexibility, rapid deployment capability, and emergency response capacity. This recommendation is derived not from optimizing a single metric, but from evaluating trade-offs among cost, time, risk, and sustainability within a unified system model.

During the construction phase, the Moon colony requires sustained, high-volume material delivery over an extended period. A purely space-elevator-based strategy, while offering a low unit transportation cost of approximately **500,000 USD/MT**, is fundamentally constrained by fixed throughput limits and exhibits vulnerability due to reliance on a single critical infrastructure system. Conversely, a rocket-only strategy provides operational flexibility but results in high cumulative exposure to launch-related disruptions as transport demand scales.

By contrast, the hybrid transportation strategy leverages the complementary strengths of both systems. Under our optimization framework, the hybrid approach achieves substantially lower generalized cost than a rocket-only strategy once time opportunity costs are considered, while maintaining far greater schedule flexibility than a space-elevator-only approach. At the system level, this balance enables the project to complete within approximately a century-scale horizon under realistic capacity constraints, with improved robustness under non-ideal operations.

Once the Moon colony enters steady-state operation, the dominant requirement shifts from one-time bulk construction to continuous and reliability-critical life-support logistics, among which *water security* is a first-order constraint. Our daily water model shows that external make-up demand

$w_{\text{ext}}(t)$  exhibits clustered peaks and persistence under disturbance conditions, rather than isolated spikes. Recycling and loss mechanisms buffer average demand, but tail-risk days can still generate short-term shortages if resupply is planned only for mean conditions. We therefore recommend sizing transportation make-up capacity and operational reserves based on high-quantile requirements, and deploying a dedicated safety reserve that is dispatched only under elevated disturbance risk. This design turns the water subsystem into a controllable “shock absorber” and prevents transient supply shocks from propagating into system-wide instability.

Beyond economics and schedule, we evaluate Earth-side environmental burden using a modular impact proxy and enforce a hard safety cap on the composite index  $EI$ . Our results indicate an explicit economy–environment trade-off: aggressive rocket intensification tends to increase Earth impact, while excessive reliance on a single infrastructure mode creates fragility. The recommended hybrid architecture remains robust under the  $EI \leq EI_{\text{max}}$  constraint by using the space elevator as the primary baseload channel (reducing repetitive atmospheric impacts) and reserving rockets as flexible peak-load and contingency capacity. We recommend adopting this safety-constrained Pareto decision protocol in planning, so that cost minimization is pursued only within an explicitly “safe” environmental region.

In conclusion, our analysis shows that a hybrid transportation architecture combining space elevators and rockets provides a robust, scalable, and sustainable pathway for building and maintaining a 100,000-person Moon colony. Rather than pursuing extreme optimization along a single dimension, this strategy achieves a practical balance across multiple objectives under realistic physical, operational, water-security, and environmental constraints. We respectfully submit this recommendation for your consideration and hope it contributes constructively to the Agencys strategic planning efforts.

Sincerely,  
Team 2605902

## 8 Strengths and weaknesses

### 8.1 Strengths

- **Unified and comparable formulation.** A single framework couples  $Z_{\text{economic}}$ ,  $\lambda T$ , and (Q2)  $\mathbb{E}[L_{\text{fail}} + L_{\text{down}}]$ , enabling consistent cross-scenario comparisons (SE-only / R-only / hybrid).
- **Traceable modular pipeline.** Parameter input  $\rightarrow$  preprocessing ( $\phi_i \mapsto m_i$ )  $\rightarrow$  MINLP  $\rightarrow$  solution  $\rightarrow$  (Q2) Monte Carlo evaluation; each module has explicit inputs/outputs.
- **Captures first-order techno-economics.** Fixed O&M costs, capacity constraints ( $K_i, Cap_{GH}$ ), and rocket learning effects via  $N_{\text{total}} = \sum_{t,i} x_{i,t}^R$  provide interpretable long-horizon trade-offs.
- **Architecture insight.** Scenario A/B/C structurally exposes baseload–peaking complementarity and single-mode vulnerability.

### 8.2 Weaknesses

- **Risk handled mainly by post-solve evaluation.** Q2 relies on an optimize-then-simulate loop, which can deviate from a true stochastic optimum.

- **Coarse temporal and operational detail.** Annual aggregation and single completion time  $T$  omit intra-year cadence, repair logistics, buffering, and recourse decisions.
- **Scalability.** MINLP complexity grows with horizon and binary variables; Q2 adds  $N_{\text{sim}}$  simulations.

### 8.3 Potential improvements with future technical breakthroughs

- **Integrated stochastic optimization.** Use SAA with decomposition (e.g., L-shaped/Benders) to optimize  $\mathbb{E}[Z]$  directly rather than only estimating it.
- **Reliability-guaranteed formulations.** Apply chance constraints or distributionally robust optimization to bound delivery shortfall risk (e.g.,  $\mathbb{P}(\text{deliver} \geq M_{\text{total}}) \geq 1 - \epsilon$ ).
- **Multi-stage recourse and buffering.** Add backlog/inventory states and adaptive reallocation to represent disruption response in hybrid operations.
- **Higher-fidelity surrogates.** Replace simplified mappings (payload, downtime) with fast physics/digital-twin surrogates while retaining tractability.

## 9 Conclusion

Demand forecasting and long-horizon lunar logistics planning are intrinsically complex, as they require the integration of multi-source uncertainties, evolving capacities, and competing objectives. Within this context, our unified modeling framework provides a coherent and reproducible pipeline from parameterization and preprocessing to optimization and (when needed) risk-aware evaluation.

Empirically, the model yields demand predictions that are highly accurate, with estimates remaining close to observed (ground-truth) values, which supports its practical validity for planning and benchmarking. Moreover, by explicitly distinguishing deterministic (Q1) and risk-aware (Q2) formulations and enabling structured scenario comparisons (SE-only, R-only, hybrid), the framework offers both quantitative performance metrics and clear mechanistic insights (e.g., baseload–peaking complementarities and robustness from substitutability).

We believe this work can serve as a useful methodological reference for future lunar transportation studies and for broader infrastructure planning problems under uncertainty. We also welcome constructive feedback and extensions, particularly on improving uncertainty characterization, operational fidelity, and computational scalability.

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## 10 Report on Use of AI Tools

In the preparation of this report, an AI language model (ChatGPT, developed by OpenAI) was used as an auxiliary tool. The AI tool was employed solely for language editing, LaTeX formatting assistance, and minor structural refinement of explanations based on content explicitly provided by the authors.

All modeling assumptions, mathematical formulations, parameter selections, computational implementations, simulations, data analyses, and conclusions were independently developed and verified by the team members. The AI tool did not generate original scientific insights, optimization models, algorithms, numerical results, or problem-specific reasoning.

The final responsibility for the accuracy, originality, and integrity of this work rests entirely with the authors.